

Indian Statistical Institute
Second Semester Mid-Sem Examination 2005-2006
B.Math.(Hons.) II Year
Discrete Mathematics & Number Theory
Time: 3 hrs Date:27-02-06 Max. Marks : 100

Answer all questions. Your answers should be complete.

1. Show that a $t - (v, k, \lambda)$ design exists, $t \geq 2$, only if $\binom{k-s}{t-s}$ divides $\lambda \binom{v-s}{t-s}$ for each $s \leq t$. [8]
2. Find the number of squares in the fields of orders 729 and 4096. [10]
3. Show that there is a unique projective plane of order 2. [15]
4. Show that there exists no projective plane of order 6. [15]
5. Let $D = (X, \mathcal{B})$ be a symmetric extendable 2-design and $B \in \mathcal{B}$. Show that $(X \setminus B, \{A : A \in \mathcal{B} \text{ and } A \cap B = \emptyset\})$ is a 2-design. Compute the number of its blocks. [12]
6. Define a symmetric 2-design. Show that any two blocks of it intersect in a constant number of points. [15]
7. Calculate the weight enumerator polynomial of the binary code generated by the incidence matrix of a projective plane of order 2. [15]
8. Determine the natural numbers n for which a projective plane of order n admits an extension. [10]

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